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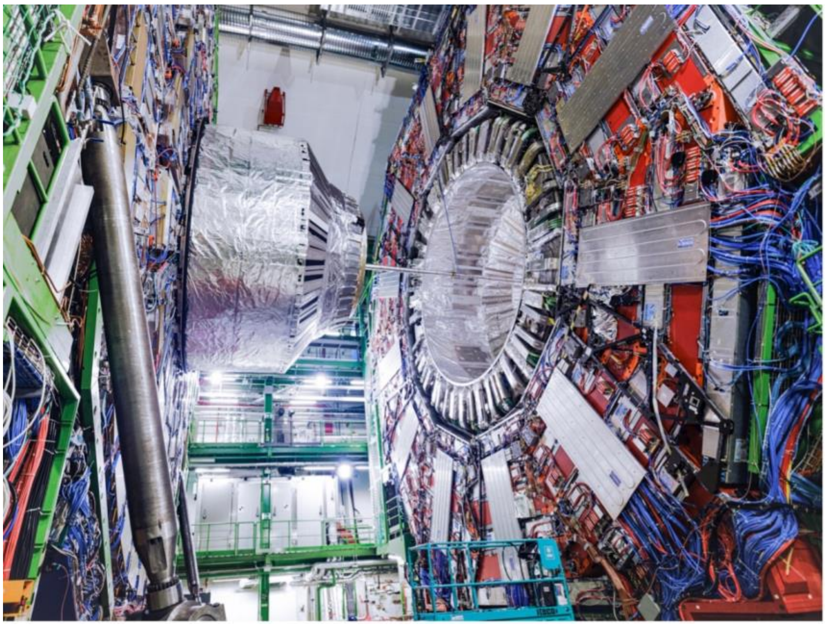
News from Moriond: The CMS experiment at CERN measures a key parameter of the Standard Model

With this measurement the LHC is again demonstrating its ability to provide very high-precision measurements and bringing new insights into an old mystery



The CMS experiment at CERN. (Image : CERN)

At the annual Rencontres de Moriond conference, the CMS collaboration presented a measurement of the effective leptonic electroweak mixing angle. The result is the most precise measurement performed at a hadron collider to date and is in good agreement with the prediction from the Standard Model.

The Standard Model of Particle Physics is the most precise description to date of particles and their interactions. Precise measurements of its parameters, combined with precise theoretical calculations, yield spectacular predictive power that allows phenomena to be determined even before they are directly observed. In this way, the Model successfully constrained the masses of the W and Z bosons (discovered at CERN in 1983), of the top quark (discovered at Fermilab in 1995) and, most recently, of the Higgs boson (discovered at CERN in 2012). Once these particles had been discovered, these predictions became consistency checks for the Model, allowing physicists to explore the limits of the theory's validity. At the same time, precision measurements of the properties of these particles are a powerful tool for searching for new phenomena beyond the Standard Model – so-called "new physics" - since new phenomena would manifest themselves as discrepancies between various measured and calculated quantities.

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A Word from Enrica Porcari

IT: interactions, innovations, impact

In the logical sequence from collisions to a scientific breakthrough, computing comes last. But definitely not least. At CERN, computing is a distributed effort. Most of the activities that are necessary for research work to happen – such as data storage, data sharing, data access and security, data architecture – are directly designed and managed within the CERN IT department. However, huge expertise in computing is also to be found in the scientific collaborations. Interactions between experts, regardless of which department or collaboration they come from, are continuous, and successful projects are often created and managed by mixed teams.

A recent example is the ALICE O2 project. ALICE's graphics processing unit (GPU)-based online data processing has the highest data recording requirement of all experiments. To address this challenge, experts from the IT department and members of the ALICE collaboration jointly developed a wholly novel approach whereby data from ALICE is sent directly to the CERN Data Centre in Meyrin, five kilometres away, through high-speed fibre-optic links. The O2 storage system is designed to be cost-efficient and highly redundant to keep data accessible even in the event of unexpected disruptions or hardware failures. The large-scale storage solution demonstrated its effectiveness last autumn, during the LHC ion run, and remains the largest disk storage system ever established in the Data Centre.

The Organization's data management strategy relies on the high standards of the two CERN Data Centres – one in Meyrin and the second, recently inaugurated, in Prévessin – and goes much beyond the CERN campus. Thanks to its network of around 160 centres distributed across 40 countries, the Worldwide LHC Computing Grid (WLCG) makes the data deluge produced by the experiments available to the wide community of scientists distributed in all the collaborating institutes around the world. This would not be possible without the thousands of innovative solutions that the community has developed over the years.

Like for the Laboratory's other activities, collaboration is also key for the IT department. It is here that the innovative public–private partnership of CERN openlab was born, over 20 years ago. Its new phase, which kicked off just a couple of weeks ago, is designed to attract new collaborations and to keep CERN openlab at the forefront of new technologies, in order to anticipate the needs of the research field and meet future challenges, including data storage at the exascale, development of innovative algorithms based on artificial intelligence and quantum computing.

We strive constantly for innovation. Here, the best talents that the Organization attracts work on technologies that will shape the future of computing, not only for science but also for society. The Quantum Technology Initiative, founded in 2018 to establish a comprehensive R&D programme in the field of quantum technologies, has evolved to also encompass a societal arm, the newly established Open Quantum Institute. The Institute will foster pioneering technological solutions that will benefit society while helping to avoid the creation of a new technological gap in our world.

We believe that the only way to shape the future is to do it in a sustainable and inclusive way. Access to science and technology must remain open to all, the cutting-edge tools that we develop must be shared with society, and we have to keep a keen eye on their environmental impact. To fulfil our challenging mission and to carry out our impactful projects, we will keep investing in attracting the best talents who believe in and share our objectives.

Enrica Porcari
Head of the IT department

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The electroweak mixing angle is a key element of these consistency checks. It is a fundamental parameter of the Standard Model, determining how the unified electroweak interaction gave rise to the electromagnetic and weak interactions through a process known as electroweak symmetry breaking. At the same time, it mathematically ties together the masses of the W and Z bosons that transmit the weak interaction. So, measurements of the W, the Z or the mixing angle provide a good experimental cross-check of the Model.

The two most precise measurements of the weak mixing angle were performed by experiments at the CERN LEP collider and by the SLD experiment at the Stanford Linear Accelerator Center (SLAC). The values disagree with each other, which had puzzled physicists for over a decade. The new result is in good agreement with the Standard Model prediction and is a step towards resolving the discrepancy between the latter and the LEP and SLD measurements.

"This result shows that precision physics can be carried out at hadron colliders," says Patricia McBride, CMS spokesperson. "The analysis had to handle the challenging environment of LHC Run 2, with an average of 35 simultaneous proton-proton collisions. This paves the way for more precision physics at the High-Luminosity LHC, where five times more proton pairs will be colliding simultaneously."

Precision tests of the Standard Model parameters are the legacy of electron-positron colliders, such as CERN's LEP, which operated until the year 2000 in the tunnel that now houses the LHC. Electron-positron collisions provide a perfect clean environment for such high-precision measurements. Proton-proton collisions in the LHC are more challenging for this kind of studies, even though the ATLAS, CMS and LHCb experiments have already provided a plethora of new ultra-precise measurements. The challenge is mainly due to huge backgrounds from other physics processes than the one being studied and to the fact that protons, unlike electrons, are not elementary particles. For this new result, reaching a precision similar to that of an electron-positron collider seemed like an impossible task, but it has now been achieved.

The measurement presented by CMS uses a sample of proton-proton collisions collected from 2016 to 2018 at a centre-of-mass energy of 13 TeV and corresponding to a total integrated luminosity of 137 fb^{-1}, meaning about 11 000 million million collisions!

The mixing angle is obtained through an analysis of angular distributions in collisions where pairs of electrons or muons are produced. This is the most precise measurement performed at a hadron collider to date, improving on previous measurements from ATLAS, CMS and LHCb.

Read more:

- CMS Physics Analysis Summary
- CMS Physics Briefing

Accelerator Report: The LHC is well ahead of schedule

Almost the whole accelerator complex is now in "physics mode", routinely delivering the various types of beam to the different physics facilities and experiments. Notably, the intensity ramp-up in the LHC is progressing remarkably well.

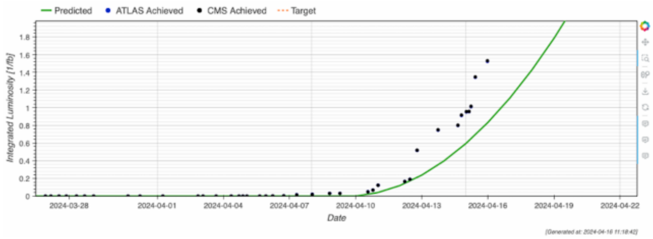
In particular, I am happy to start this report with the good news that, thanks to the excellent availability of the accelerator complex and the hard work of the LHC teams and experts, the LHC is now 12 days ahead of schedule, yielding a direct gain of integrated luminosity and thus physics and boding well for the 2024 run.

The first stable beams of 2024 in the LHC were initially scheduled for 8 April, but the teams working on the LHC beam commissioning managed to be ready earlier and declared first stable beams at 18.25 on 5 April, three days ahead of the official schedule. The first stable beams also mark the start of a period of intensity ramp-up interleaved with the completion of the final commissioning steps.

These final steps include the scrubbing of the LHC vacuum chamber to reduce the production of electron clouds that negatively impact the beam quality and put a strain on the cryogenics system. Usually, the scrubbing lasts two days, but this year an extra day was added since a new injection kicker and two TDIS (target dump injection systems) were installed during the YETS (the new TDIS replace the ones at Points 2 and 8 that suffered vacuum leaks in 2023). The scrubbing run was nevertheless completed in only 36 hours, resulting in another gain in the schedule.

The LHC availability during the recent intensity ramp-up was 85%, including stable beams for about 35% of the time, and the experts very efficiently signed off the checklists at each intensity step. This is why we are now about 12 days ahead of schedule, colliding beams of 1200 bunches and already producing a meaningful level of luminosity for physics. The next step is 1800 bunches, which, if all goes well, might be achieved before the end of this week.

Meanwhile, the injectors are providing the experiment facilities with beams for physics. The PS was the first to routinely provide beams for physics to the East Area, on 22 March, and n_TOF followed suit on 25 March. ISOLDE, located behind the PS Booster, started physics on 8 April. The SPS fixed-target physics in the North Area started on 10 April. On 15 April, the AWAKE facility located behind the SPS started the first of five two-week proton runs scheduled for 2024. The next in line is the Antimatter factory: the AD and ELENA decelerators should start providing the experiments with antiprotons for physics on 22 April.



On Tuesday, 16 April, at the end of the afternoon, the first 1.5 fb^{-1} of integrated luminosity was collected. More than 90 fb^{-1} are expected for 2024. (Image: CERN)

The 2024 run has been extended by four weeks, until 25 November, for the LHC, and by five weeks, until 2 December, for the injectors. The YETS will start later this year, which will allow more physics to be done in 2024.

Rende Steerenberg

CERN70: Cutting-edge computing

Part 7 of the CERN70 series. Paolo Zanella came to the CERN computing group in 1962, just a few years after the first computer had arrived

CERN's first computer, a huge vacuum-tube Ferranti Mercury, was installed in 1958. It represented the first stage in the evolution of digital computing at CERN.

The next big step came in 1965, when the first supercomputer arrived: a CDC 6600 designed by computer pioneer Seymour Cray. This transistorised sub-microsecond machine was the first real "number cruncher". CERN was among the first to adopt these new technologies, which required a great deal of effort to implement because of hardware instabilities.

In 1972, CERN installed the CDC 7600, then the most powerful machine on the market, five times faster than the 6600. Moving from hardware to software, CERN stepped up its efforts to adopt cutting-edge information technology (IT). These two CDC machines played a leading role in computing for high-energy physics. For many years, no other machine was as advanced and as fast as the CDC 7600, which was finally disconnected in 1984. In 12 years of service, it had processed a huge amount of work from thousands of users.

In parallel, from 1973 to 1980, CERN built highly advanced computer systems to control the accelerators. They made the Proton Synchrotron into a flexible workhorse, time-shared by its customers the Super Proton Synchrotron and later the Large Electron Positron collider. Here CERN's engineers implemented networks, distributed computer systems, human-computer interfaces with touchscreens, trackballs and more. The trend shifted to a decentralised system with many smaller computers, rather than one large central system that dominated computing.

The tape units of CERN's first supercomputer, a CDC 6600, which arrived in 1965. (Image: CERN)

The huge scientific experiments that started to appear in the 1980s, as well as those that now take data at the Large Hadron Collider, would not have been possible without a substantial human and material investment at CERN in the information processing and communication technologies. An extremely rewarding by-product has been the invention of the World Wide Web at CERN in the early 1990s, which has made a fundamental impact on science and society.

Recollections

Paolo Zanella came to the CERN computing group in 1962, just a few years after the first computer had arrived. He was later head of the Data Handling Division from 1976 to 1989, before becoming professor at the University of Geneva.

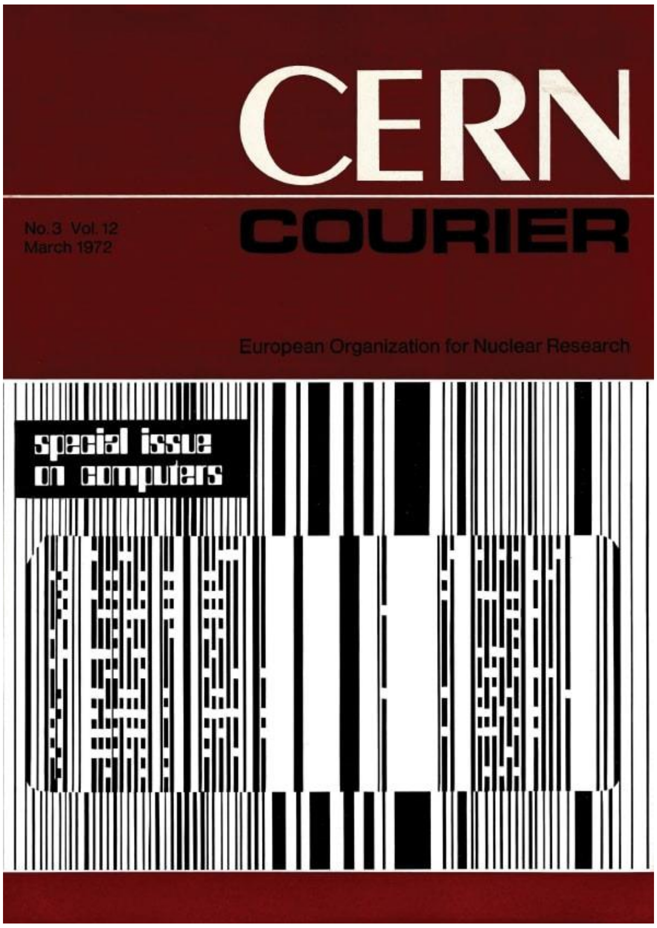
"When I came to CERN, my job was not only to find problems and solve them, but also to try and convince the physicists that computers were something useful. Only a few days after I arrived, Lew Kowarski, then leader of the Data Handling Division, sent me to see Carlo Rubbia, "a future Nobel Prize winner" in Kowarski's own words, to persuade him to use the recently installed IBM 709. Rubbia's welcome was that physicists did not need it and did not need me. But then he called me back and said that we should try this and try that – in the end, he was one of the first to adopt computer technology and used it widely and successfully in his experiments throughout his brilliant experimental work at CERN and elsewhere.

When I was appointed division leader in 1976, it was with an assignment to reverse trends. The division was in crisis. We had powerful but unreliable computers and some software disasters. The physicists were furious. But a few years later, they were pleased about the changes and the division had earned respect.

The Ferranti Mercury computer was CERN's first computer. It was installed in Building 2 on 30 June 1958. (Image : CERN)

We provided excellent service and at the same time we did drive development in the field. However, the relationship with the researchers has never been easy. When we asked for money to develop networks in the 1970s they "could not see why two computers would be interested in talking to each other", and later the cabling of buildings and the advent of workstations was strongly opposed. In the end, IT invaded the experimental floors, the engineers' and administrators' offices, and just at the end of my professional life at CERN, when the World Wide Web was invented, we were recognised as one of the best and most advanced IT environments in scientific research worldwide.

I learned a lot in my training at CERN – and not only technically. Here I got the courage to take on the impossible and to win. I lived during one of the most interesting half-centuries of particle physics. At CERN, I learned a few basic questions that work universally: What is it for? Why do we need it? Will it work? And if you make something useful that works, it will surely be praised and adopted.



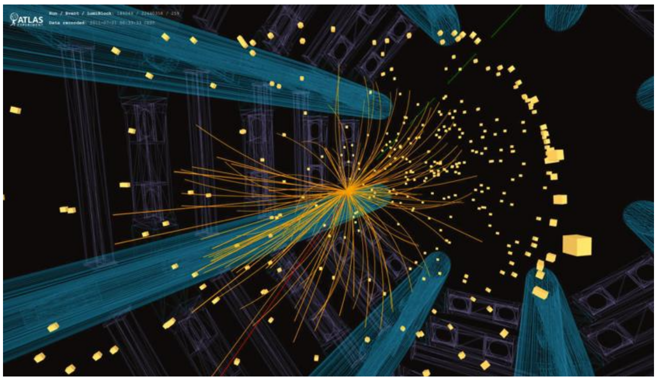
The March 1972 CERN Courier was a special issue on computers, beginning with the article "Computers: why?" (Image: CERN)

In the 1960s, computing was there but nobody knew what to do with it. On the other side was science, and they had to come together step-by-step. Information technology is a universal machinery that can do a lot for science and for civilisation. IT and particle physics have followed very similar growth curves pushing and changing each other. I have been very lucky to live most of my life at the frontier between these two exciting disciplines."

This interview is adapted from the 2004 book "Infinitely CERN", published to celebrate CERN's 50th anniversary. More information about CERN's computing history is available on the CERN IT department website.

News from Moriond: ATLAS provides first measurement of the W-boson width at the LHC

The measurement is the most precise yet made by a single experiment



View of an ATLAS collision event in which a candidate W boson decays into a muon and a neutrino. The reconstructed tracks of the charged particles in the inner part of the ATLAS detector are shown as orange lines. The energy deposits in the detector's calorimeters are shown as yellow boxes. The identified muon is shown as a red line. The missing transverse momentum associated with the neutrino is shown as a green dashed line. (Image: ATLAS/CERN)

The discovery of the Higgs boson in 2012 slotted in the final missing piece of the Standard Model puzzle. Yet, it left lingering questions. What lies beyond this framework? Where are the new phenomena that would solve the Universe's remaining mysteries, such as the nature of dark matter and the origin of matter–antimatter asymmetry?

One parameter that may hold clues about new physics phenomena is the "width" of the W boson, the electrically charged carrier of the weak force. A particle's width is directly related to its lifetime and describes how it decays to other particles. If the W boson decays in unexpected ways, such as into yet-to-be-discovered new particles, these would influence the measured width. As its value is precisely predicted by the Standard Model based on the strength of the charged weak force and the mass of the W boson (along with smaller quantum effects), any significant deviation from the prediction would indicate the presence of unaccounted phenomena.

In a new study, the ATLAS collaboration measured the W-boson width at the Large Hadron Collider (LHC) for the first time. The W-boson width had previously been measured at CERN's Large Electron–Positron (LEP) collider and Fermilab's Tevatron collider, yielding an average value of 2085 ± 42 million electronvolts (MeV), consistent with the Standard-Model prediction of 2088 ± 1 MeV. Using proton–proton collision data at an energy of 7 TeV collected during Run 1 of the LHC, ATLAS measured the W-boson width as 2202 ± 47 MeV. This is the most precise measurement to date made by a single experiment, and — while a bit larger — it is consistent with the Standard-Model prediction to within 2.5 standard deviations.

This remarkable result was achieved by performing a detailed particle-momentum analysis of decays of the W boson into an electron or a muon and their corresponding neutrino, which goes undetected but leaves a signature of missing energy in the collision event (see image above). This required physicists to precisely calibrate the ATLAS detector's response to these particles in terms of efficiency, energy and momentum, taking contributions from background processes into account.

However, achieving such high precision also requires the confluence of several high-precision results. For instance, an accurate understanding of W-boson production in proton–proton collisions was essential, and researchers relied on a combination of theoretical predictions validated by various measurements of W and Z boson properties. Also crucial to this measurement is the knowledge of the inner structure of the proton, which is described in parton distribution functions. ATLAS physicists incorporated and tested parton distribution functions derived by global research groups from fits to data from a wide range of particle physics experiments.

The ATLAS collaboration measured the W-boson width simultaneously with the W-boson mass using a statistical method that allowed part of the parameters quantifying uncertainties to be directly constrained from the measured data, thus improving the measurement's precision. The updated measurement of the W-boson mass is 80367 ± 16 MeV, which improves on and supersedes the previous ATLAS measurement using the same dataset. The measured values of both the mass and the width are consistent with the Standard-Model predictions.

Future measurements of the W-boson width and mass using larger ATLAS datasets are expected to reduce the statistical and experimental uncertainties. Concurrently, advancements in theoretical predictions and a more refined understanding of parton distribution functions will help to reduce the theoretical uncertainties. As their measurements become ever more precise, physicists will be able to conduct yet more stringent tests of the Standard Model and probe for new particles and forces.

ATLAS collaboration

News from Moriond: Searching for new asymmetry between matter and antimatter

LHCb has conducted a new search for matter–antimatter asymmetry using its full data sets from the first and second runs of the LHC

Once a particle of matter, always a particle of matter. Or not. Thanks to a quirk of quantum physics, four known particles made up of two different quarks – such as the electrically neutral D meson composed of a charm quark and an up antiquark – can spontaneously oscillate into their antimatter partners and vice versa.

At a seminar held recently at CERN, the LHCb collaboration at the Large Hadron Collider (LHC) presented the results of its latest search for matter–antimatter asymmetry in the oscillation of the neutral D meson, which, if found, could help shed light on the mysterious matter–antimatter imbalance in the Universe.

The weak force of the Standard Model of particle physics induces an asymmetry between matter and antimatter, known as CP violation, in particles containing quarks. However, these sources of CP violation are difficult to study and are insufficient to explain the matter–antimatter imbalance in the Universe, leading physicists to both search for new sources and to study the known ones better than ever before.

In their latest endeavour, the LHCb researchers have rolled up their sleeves to measure with unprecedented precision a set of parameters that determine the matter–antimatter oscillation of the neutral D meson and enable the search for the hitherto unobserved but predicted CP violation in the oscillation.

The collaboration had previously measured the same set of parameters, which are linked to the decay of the neutral D meson into a positively charged kaon and a negatively charged pion, using its full data set from Run 1 of the LHC and a partial data set from Run 2. This time around, the team analysed the full Run-2 data set and, by combining the result with that of its previous analysis, excluding the partial Run-2 data set, it obtained the most precise measurements of the parameters to date – the overall measurement uncertainty is 1.6 times smaller than the smallest uncertainty achieved before by LHCb.

The results are consistent with previous studies, confirming the matter–antimatter oscillation of the neutral D meson and showing no evidence of CP violation in the oscillation. The findings call for future analyses of this and other decays of the neutral D meson using data from the third run of the LHC and its planned upgrade, the High-Luminosity LHC.

Other neutral D meson decays of interest include the decay into a pair of two kaons or two pions, in which LHCb researchers observed CP violation in particles containing charm quarks for the first time, and the decay into a neutral kaon and a pair of pions, with which LHCb clocked the speed of the particle's matter–antimatter oscillation. No avenue should be left unexplored in the search for clues to the matter–antimatter imbalance in the Universe and other cosmic mysteries.

Find out more on the LHCb website.

Ana Lopes

News from Moriond: FASER measures high-energy neutrino interaction strength

The interaction strength of neutrinos had never previously been measured in this energy range

Operating at CERN's Large Hadron Collider (LHC) since 2022, the FASER experiment is designed to search for extremely weakly interacting particles. Such particles are predicted by many theories beyond the Standard Model that are attempting to solve outstanding problems in physics such as the nature of dark matter and the matter-antimatter imbalance in the Universe. Another goal of the experiment is to study interactions of high-energy neutrinos produced in the LHC collisions, particles that are nearly impossible to detect in the four big LHC experiments. Last week, at the annual Rencontres de Moriond conference, the FASER collaboration presented a measurement of the interaction strength, or "cross section", of electron neutrinos (ν_e) and muon neutrinos (ν_μ). This is the first time such a measurement has been made at a particle collider. Measurements of this kind can provide important insights across different aspects of physics, from understanding the production of "forward" particles in the LHC collisions and improving our understanding of the structure of the proton to interpreting measurements of high-energy neutrinos from astrophysical sources performed by neutrino-telescope experiments.

FASER is located in a side tunnel of the LHC accelerator, 480 metres away from the ATLAS detector collision point. At that location, the LHC beam is already nearly 10 metres away, bending away on its circular 27-kilometre path. This is a unique location for studying weakly interacting particles produced in the LHC collisions. Charged particles produced in the collisions are deflected by the LHC magnets. Most neutral particles are stopped by the hundreds of metres of rock between FASER and ATLAS. Only very weakly interacting neutral particles like neutrinos are expected to continue straight on and reach the location where the detector is installed.

The probability of a neutrino interacting with matter is very small, but not zero. The type of interaction that FASER is sensitive to is where a neutrino interacts with a proton or a neutron inside the detector. In this interaction, the neutrino transforms into a charged "lepton" of the same family – an electron in the case of an ν_e, and a muon in the case of a ν_μ – which is visible in the detector. If the energy of the neutrino is high, several other particles are also produced in the collision.

The detector used to perform the measurement consists of 730 interleaved tungsten plates and photographic emulsion plates. The emulsion was exposed during the period from 26 July to 13 September 2022 and then chemically developed and analysed in search of charged particle tracks. Candidates for neutrino interactions were identified by looking for clusters of tracks that could be traced back to a single vertex. One of these tracks then had to be identified as a high-energy electron or muon.

In total, four candidates for an ve interaction and eight candidates for a ν_μ interaction have been found. The four ν_e candidates represent the first direct observation of electron neutrinos produced at a collider. The observations can be interpreted as measurements of neutrino interaction cross sections, yielding (1.2+0.9−0.8) $\times 10^{-38}$ cm^2 GeV^{-1} in the case of the ν_e and $(0.5 \pm 0.2) \times 10^{-38}$ cm^2 GeV^{-1} in the case of the ν_μ. The energies of the neutrinos were found to be in a range between 500 and 1700 GeV. No measurement of the neutrino interaction cross section had previously been made at energies above 300 GeV in the case of the ve and between 400 GeV and 6 TeV in the case of the ν_μ.

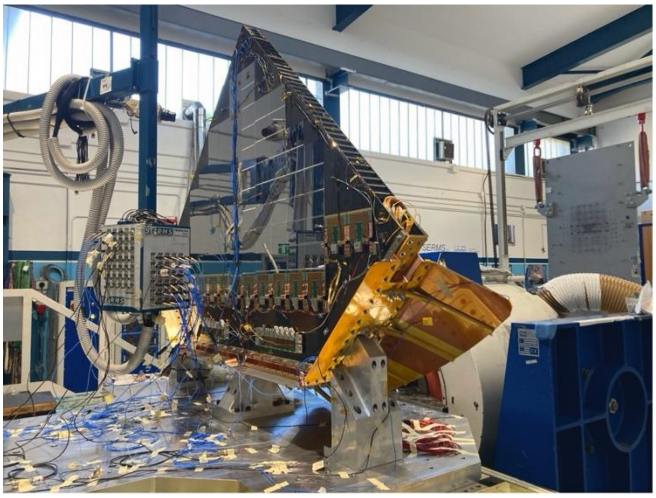
The results obtained by FASER are consistent with expectations and demonstrate the ability of FASER to make neutrino cross-section measurements at the LHC. With the full LHC Run 3 data, 200 times more neutrino events will be detected, allowing much more precise measurements.

Read more in the FASER publication.

Piotr Traczyk

AMS's second new life

After more than a decade in space, the AMS detector on the ISS is due for another upgrade. With its augmented potential for discovery, it will be able to investigate dark-matter-like signatures from cosmic rays



A quarter plate for the new tracking layer goes through a vibrational test at INFN Perugia (Image: Corrado Gargiulo/INFN)

In 2011, the Alpha Magnetic Spectrometer (AMS) was installed on the International Space Station (ISS). Since then, it has recorded more than 200 billion cosmic ray events and, while most of their sources are known, a few signatures in the data could point to dark matter. The detector's latest upgrade will enable scientists to investigate this further.

AMS collects cosmic ray particles that reach Earth coming either directly from the Sun or from far-away sources such as stars ending in supernovae or black holes. Most of the cosmic rays that AMS detects are protons, but heavy nuclei like iron or silicon also reach the detector. However, one signature is particularly intriguing. AMS has detected an unusually high flux of positrons – the antimatter partners of electrons. Positrons and other antimatter particles are rare in the Universe and hence not expected to be seen in the observed data at the strength found by AMS. Their origin is not yet confirmed; they could come from pulsars (fast rotating remnants of stars that emit regular signals), a yet-unknown astrophysical source or dark matter. The observed positron flux fits very well with dark matter models. But in order to investigate this more accurately, the AMS collaboration is now working on refurbishing the detector.

The main upgrade will be a new detector layer with a higher number of silicon strips that will increase the acceptance of recording infalling particles by 300%. "By 2030, AMS will extend the energy range of the positron flux and reduce the error by a factor of two compared with current data," says AMS spokesperson Sam Ting (MIT). This will allow the detector to investigate the positron signature even further.

A second important addition will be three new radiative surfaces. Because AMS is exposed to direct sunlight, it was painted white to reflect excess heat and remain at operational temperatures. After 13 years in the demanding conditions of space, the paint has degraded and, to compensate for this, the new radiators will keep AMS cool again.

Currently, all the parts of the new upgrade, including electronics and hardware, are being built as "validation" and "qualification" models. If they pass all the tests happening at CERN, INFN Perugia and IABG in Germany, the final flight model will go into production. Astronauts are already training with the prototypes in space-like environments on Earth. In 2026, when the upgrade is launched, the astronauts will mount the new detector parts onto AMS during spacewalks. "Everything is going very, very fast," says chief engineer Corrado Gargiulo (CERN). "This is a requirement, otherwise we arrive too late at the ISS for the upgrade to make sense." Indeed, the mission now has an end date. NASA has scheduled the deorbiting of the ISS for 2030 and, until then, AMS will have plenty of cosmic ray events to record to explore the positron signature.

Sanje Fenkart

Mitigating the environmental impact of CERN procurement

A comprehensive strategy to minimise emissions arising from the Organization's purchases

Every year, CERN spends some 500 MCHF on goods and services to build, maintain and operate its infrastructure to fulfil its scientific objectives. These purchases not only come at a financial cost, but also have an impact on the environment through the indirect emissions arising from their procurement. In 2023, CERN reported its procurement-related indirect emissions in the CERN Environment Report for the first time. These amounted to 98 030 tCO2e and 104 974 tCO2e in 2021 and 2022 respectively. To put this in context, this represents more than 90% of CERN's total indirect emissions, the rest being attributed to personnel mobility, duty travel and catering, and just over 30% of CERN's total emissions.

CERN strives to be a model for environmentally responsible research by taking action on its most impactful domains, including energy and water consumption and emissions, and setting objectives to minimise its environmental footprint. Adopting measures to positively influence procurement-related emissions is a priority for which a comprehensive strategy has been set out that will commit CERN, its suppliers and each and every one of us to making conscious decisions when purchasing goods or services.

Underpinning this strategy, the Environmentally Responsible Procurement Policy was approved by the Enlarged Directorate in June 2023. Anchored in the principle of embedding environmental responsibility where appropriate throughout all phases of the procurement process, the Policy commits the Organization to environmentally responsible procurement and to achieving sustainable results both internally and throughout its supply chains, integrating relevant best practices in its processes, measuring their impact, and communicating with and raising the awareness of all stakeholders.

In December 2023, the Enlarged Directorate approved the implementation of the Policy, effective from 1 January 2024. This entails a one-year kick-off phase to identify suitable areas for policy implementation, including a comprehensive awareness-raising programme with tailored training for technical officers and workshops for the departments focusing on their purchasing activities.

Additionally, pilot projects will help evaluate the integration of environmental criteria into market surveys and invitations to tender. Procurement officers will have access to a supplier sustainability due diligence tool and guidelines outlining best practices. These resources will equip them with the knowledge they need to assess suppliers based on their sustainability efforts.

Furthermore, a supplier engagement programme will be launched in order to foster discussions on sustainability within our supply chains, aiming to collaborate with and encourage suppliers to adopt sustainable practices.

Overall, this comprehensive implementation plan is designed to ensure a smooth transition towards policy compliance and create a sustainable framework for all stakeholders involved. Successful implementation will depend on all actors in CERN's supply chains challenging our choices and decisions, from CERN's IPT department, to CERN personnel involved in purchasing, to the suppliers themselves spanning our 23 Member and 11 Associate Member States, while continuing to strive for balanced returns.

According to Chris Hartley, Head of the IPT Department: "It is of great importance that we have established an Environmentally Responsible Procurement Policy for CERN. All CERN stakeholders want to see CERN continue to minimise its environmental impact. This Policy, underpinned by our progressive commitment to responsible sourcing, waste reduction and supplier engagement, will contribute to a more sustainable future."

IPT department

Work Well Feel Well: Get rid of your strains

Part four of the Work Well Feel Well series looks at ways to release tension

Release the pressure

Many of us would like to leave our stress behind, but we can find it **difficult to let go**. Perhaps we fear that by doing so, we won't have the strength to deal with the demands of everyday life. This may well be the case when we are **tired** or **suffering from too much stress**.

Much like a pressure cooker, it can be unwise to "open the lid in one go" and relieve all the accumulated pressure without first preparing. But we can release pent-up pressure in stages, progressively. This can help to unburden us and give us more energy.

The **body** sends out **warning signs** when we have been under stress or when we hold on to an emotion at the expense of our health:

- Shortness of breath or shallow breathing.
- Clenched jaw.
- Tight and dry throat.
- Tension in the shoulders and neck.
- Knot in the stomach or nervous tension in the solar plexus.

These warning signs can drain our energy and wear us out in the long term. Restoring our body's equilibrium is an essential step in managing our stress. Three physical fundamentals are particularly efficient in **helping** to get rid of **stress**:

1. Breathing, especially exhaling
2. Body movements
3. Vocal exercises

In a few simple gestures, our stress can give way to calm. Freeing our mind and body gives us the inner strength that we need to face up to pressures, challenges or difficulties.

Take action

As part of the "**Efficiency and caring at work**" campaign, the Work Well Feel Well website now offers **useful resources** that can be downloaded, including **exercises** on **breathing** and on discovering **safety valves** to help release pressure. In addition:

- On 4 June, the **CERN psychologists** are hosting **lunchtime** sessions in English and French on useful **techniques to manage stress** in our daily lives.
- CERN HR Learning and Development also provides training courses to help boost resilience and tackle stress:
  - For supervisors: "Keeping Pressure Positive: A leadership perspective"
  - For teams: "Building Team Resilience through positive Stress Management"
  - For individuals: "Balancing Performance and Pressure"

This is part four of a 12-part Work Well Feel Well series, with articles published every two months.

HR department

CERN and the Swiss Arts Council announce the artists selected for the sixth edition of Connect

Arts at CERN, in collaboration with the Swiss Arts Council Pro Helvetia, today announced Robin Meier Wiratunga and Vimala Pons as the artists selected for the sixth edition of Connect

Connect is an art residency programme launched by Arts at CERN and Pro Helvetia in 2021. Dedicated to Swiss-based artists working at the intersection of science and artistic research, this research-led residency invites them to come to CERN to explore ideas and develop new work.

The selected duo is composed of Robin Meier Wiratunga and Vimala Pons. Meier Wiratunga is an artist and composer who seeks to understand how humans, insects and objects think. Collaborating closely with scientific researchers, Meier Wiratunga's work blends machine learning with insights from animal intelligence, creating constellation scores where musical patterns emerge as 'thinking tools'. Vimala Pons is an actress who has worked in independent and auteur cinema.

Meier Wiratunga and Pons will dedicate their residency at CERN to developing their proposal titled Guided Meditations for the End of the Universe. This project aims to delve into cosmological theories about the end of the Universe to transform them into meditative and embodied experiences. Employing an anthropological approach to engage with science, the artists intend to employ spoken voice recordings, electronic music, light environments and sonified data from particle events.

Now in its sixth edition, Connect has become a pivotal platform for Swiss-based artists to expand their artistic practice in dialogue with the field of physics and science at CERN. This collaboration framework between Arts at CERN and Pro Helvetia will continue through the next year with an iteration in Chile and India, sustaining its mission to foster interactions and dialogue between artistic and scientific communities.

"We find ourselves in a challenging yet exciting time, as we witness the emergence of a strong and vibrant infrastructure devoted to the integration of artistic activity within the sciences," says Mónica Bello, Curator and Head of Arts at CERN. "This is a transformational and promising development that presents unprecedented opportunities for cultural innovation. Connect is evidence of these dynamics, and I am proud to see that our partnership with Pro Helvetia is advancing further, bringing in new residents with different perspectives and backgrounds and strengthening the confluence of art and science in Switzerland."

"This sixth edition of Connect continues to reaffirm the significance of the interface between art, science and technology. The quality and diversity of applications received illustrate the rich interdisciplinarity inherent in this dynamic field, and the extent to which it is part of current artistic practice. We are happy to foster dialogue and innovation at this remarkable intersection and are very pleased about the ongoing collaboration with Arts at CERN," explains Philippe Bischof, Director of Pro Helvetia.

The jury of Connect was formed by Mónica Bello, Curator and Head of Arts at CERN; Giulia Bini, Head of Program and Curator of "Enter the Hyper-Scientific" at EPFL Lausanne; and Federica Martini, Head of the CCC - Critical Curatorial Cybermedia Master at HEAD Genève.

Blazing trails: CMS cavern evacuation paves the way for future safety design

How an evacuation of the CMS experimental cavern has provided invaluable human behaviour data to improve emergency preparedness for complex underground facilities

CERN strives for excellence in safety matters, with a commitment to continuous improvement in the field. Emergency preparedness is a priority for the Organization as it is a key element in its aim to protect both people participating in its activities and its installations. In this context, regular evacuation exercises of all accelerator and experimental areas are a regulatory requirement and part of the CERN-wide safety objectives.

On a warm, sunny day in February 2024, 48 people were going about their daily work in the CMS cavern, unaware that an evacuation exercise, which had been carefully planned for several months, was about to take place. Such exercises are crucial for facility users and rescue teams to gain familiarity with emergency procedures in various contexts and settings. When the alarm sounded, all 48 people reacted calmly, reaching the assembly point quickly and safely. It was a pleasing result for CMS and, apart from the important lessons learned from the exercise, additional data was gathered to improve not only evacuation procedures but also the design of installations in order to make emergency plans even more effective.

This exercise was part of a pilot collaboration between CMS Safety, the HSE Fire Safety Engineering (FSE) team and the Fire Safety Engineering division of Lund University in Sweden, which took this opportunity to maximise the usefulness of the evacuation to study human behaviour in emergency situations.

Comprising reports by undercover observers, questionnaires and footage from security cameras (used in full compliance with Operational Circular No. 11 to ensure anonymity), the data collected provides many useful insights into evacuation dynamics, occupant characteristics and perceptions of safety procedures.

This information is essential for the design of and emergency planning for subterranean experimental areas. As opposed to the design of buildings located above ground, which follows national safety standards, the design of underground areas relies extensively on computer modelling. Using various parameters, it is possible to simulate human behaviours in the event of an emergency to predict the effectiveness of a real-life evacuation.

In this pilot study, the Lund and FSE teams will use the CMS evacuation data to identify unique human behaviours observed in emergencies in complex underground environments. This will expand the current knowledge base and help build a database of specific input parameters to fine-tune and/or validate existing evacuation models.

Ultimately, this methodology will be instrumental not only to improve CERN's emergency response in the caverns, but also to influence the safety design across current and future complex facilities, at CERN and beyond.

CMS collaboration & HSE Unit

CERN donates computing equipment to South Africa

Bob Jones (left), deputy head of CERN's IT department, Mr. Curtis Singo (middle) Political and Economic Counsellor at the South Africa Embassy in Bern, and Joachim Mnich (right), CERN's director for Research and Computing, in building 133, where the computer hardware was prepared for shipment. (Image: CERN)

On 9 April 2024, a ceremony at CERN marked the donation of computing equipment to the Tshwane University of Technology in South Africa. The ceremony was attended by Mr. Curtis Singo, Political and Economic Counsellor at the South Africa Embassy in Bern, Joachim Mnich, CERN's director for Research and Computing, and Bob Jones, deputy head of CERN's IT department.

On this occasion, 21 servers and 4 network switches were sent to the Tshwane University of Technology, where the equipment will be used to support academic and research projects.

CERN regularly donates computing equipment that no longer meets its highly specific requirements but is still more than adequate for less demanding environments. To date, more than 2500 servers and 150 network switches have been donated by CERN to countries and international organisations, namely Algeria, Bulgaria, Ecuador, Egypt, Ghana, Mexico, Morocco, Nepal, Palestine, Pakistan, the Philippines, Senegal, Serbia, Jordan, Lebanon and now South Africa.

If you are a publicly funded research organisation, you can request computing equipment from CERN.

Marina Banjac

Handover at the CERN Ombud’s Office

The CERN Ombud’s Office was established in 2010 to provide the entire CERN community with support in resolving conflicts informally, in a consensual and impartial manner. Since then, several Ombuds have held the position, which is now firmly anchored at the Laboratory. On 1 May, Marie-Luce Falipou, the fifth CERN Ombud, will take up her duties. She takes over from Laure Esteveny, who has been in the role since April 2021 and is taking early retirement.
As they prepared for the handover, Laure and Marie-Luce agreed to answer the *Bulletin* team’s questions.

Laure Esteveny (left) hands the keys to the Ombud’s office to Marie-Luce Falipou, whose new role begins on 1 May. (Image: CERN)

Bulletin: Laure, what drove you to become Ombud?
Laure: I began my career as Ombud in 2021, after 35 years working in different departments of the Organization. I was attracted by the human side of the role, and I haven’t been disappointed. I’m extremely happy to have been able to serve out my career as CERN Ombud. It’s very rewarding to help people to overcome a conflict.

Bulletin: What do you need to succeed in this role?
Laure: When they take up their duties, all new CERN Ombuds follow the training courses run by the International Ombuds Association (IAO) and also receive training in mediation. This is clearly essential, but the skills I acquired throughout my career at CERN have also been invaluable. You need to have good analytical skills and be very thorough to succeed in this role.
Pierre Gildemyn, my predecessor, also supported me a lot. He was always available to answer my questions and shared his own experience as Ombud with me. I, in turn, am available for Marie-Luce; I will be delighted to help her. I would also urge her to turn to the IAO for support and to all the professional ombud networks, especially that of the United Nations and Related International Organizations (UNARIO) – ombuds are very good at supporting each other.

Bulletin: Have you encountered any difficulties?
Laure: The role of Ombud is very rewarding on the human level. If I had to name a difficulty, I would say that the isolation that inevitably comes with the role is not always easy to cope with. In addition, by definition, the Ombud is only exposed to problematic situations in which people are suffering – to the “Dark Side of the Force” – which can sometimes be a heavy burden.

Bulletin: Marie-Luce, what brought you to this role?
Marie-Luce: I’ve known Pierre and Laure for a long time, and I’ve seen them thrive in the role of Ombud, for which they developed a real passion. It’s a privilege to be the next to take on this role.
I’ve spent my whole career at CERN, 35 years now, in the Human Resources department. I’ve held various positions, notably HRA (human resources adviser) for 13 years, so I’m very familiar with the workplace culture. I’ve also been trained in active listening – skills that will certainly be very useful in my new position. For me, becoming Ombud is really a natural evolution, even if the role is of course unique.

Bulletin: You take up your duties on 1 May, but you don't become Ombud overnight, I imagine?
Marie-Luce: Indeed, I'll take the necessary time to prepare myself for this new role, which is really something out of the ordinary and something with which I need to familiarise myself. I'm aware of the importance and the impact that the Ombud can have, and I'm humbled and grateful to accept this responsibility.

Bulletin: As the new Ombud, what message would you like to send to the CERN community?
Marie-Luce: I want people to know that the Ombud's office is a safe, calm place where they will be listened to in confidence and understood. The Ombud is there to serve all members of the CERN community, regardless of their role in the Organization. Questions, problems and conflicts are part and parcel of life, including in the workplace. Finding the best way to handle them can make a big difference, and that's where the Ombud can help.

Bulletin: Any final words?
Marie-Luce: I'd like to say a big thank you to Laure for sharing her experience with me and for offering me her support; it's a precious resource to have someone experienced to turn to.
Laure: Thank you to all those who placed their trust in me, and I wish Marie-Luce all the best!

Bulletin: A big thank you to you both!

Internal Communication

The Ombud is available from Monday to Friday in office B500/1-004 on the Meyrin site. To make an appointment, in person or online, contact the Ombud at ombuds@cern.ch.
More information can be found on the Ombud's website: https://ombuds.web.cern.ch

Fabiola Gianotti receives the 2024 prize from the "Fondation pour Genève"

The *Fondation pour Genève* will be presenting its 30th prize to Fabiola Gianotti, CERN Director-General, in recognition of her outstanding contribution to Geneva's international reputation.
"I am extremely honoured to receive the *Fondation pour Genève* Prize. The development of science and technology, openness, collaboration across borders and the education of young people are fundamental values at CERN, which are also deeply rooted in international Geneva. The fact that these values, which are so dear to me, are being recognised is a particularly touching moment for me," declared Fabiola Gianotti.
The award ceremony is open to everyone and will take place on **Monday 13 May 2024 at 6.30 p.m.** at the Victoria Hall in Geneva. To register, click here.
More information on the Fondation pour Genève *website.*

CERN to change name for 70th Anniversary

The Organization's new name, Network of Experiments for Research and Development in Society, encapsulates the true nature of the Laboratory in 2024

Update: Did you enjoy our April Fools' day story? CERN is indeed celebrating its 70th anniversary, but without changing its name. For the complete CERN70 anniversary events and programme of activities, visit cern.ch/cern70.

Since its inception in 1954, CERN has grown from a small group of physicists from a handful of countries to a thriving international hub for science and technology. This is why, on the occasion of the Laboratory's 70th Anniversary, the time has come for the name to be adapted to reflect its new role in society. The new name, the Network of Experiments for Research and Development in Society, will come into force on 1 October 2024, at the Laboratory's 70th birthday party.

The Organization's new logo. The correct usage of the branding can be found on the Design Guidelines website. (Image: CERN)

The acronym CERN was borne from an intergovernmental meeting of UNESCO in Paris in December 1951. This is when the first resolution concerning the establishment of a European Council for Nuclear Research (in French Conseil Européen pour la Recherche Nucléaire, or CERN) was adopted. Two months later, an agreement was signed establishing the provisional Council – and the name "CERN" stuck. However, today, our understanding of matter goes much deeper than the nucleus, and "CERN" is now widely viewed across the scientific community as an outdated and exclusionary name.

"The word "nuclear" doesn't really reflect the full breadth of scientific research we do here," says Noah Lott, Head of Rebranding at the Organization. "Network of Experiments for Research and Development in Society indicates the range of particle physics, computing, engineering and technology research that takes place at the Laboratory, as well as its impact on society."

"We are also no longer just a European organisation, as we have grown to a global community encapsulating more than 80 countries," adds Ivana Reed, spokesperson for international relations at the Laboratory. "We feel that everyone, no matter who they are, will feel accepted and proud to be associated with NERDS."

A dedicated working group – the Decision for a Unified Moniker Board – was created in 2021 to assess potential options for the new name. "NERDS is so much more memorable and inclusive than CERN," explains Wirall Geex, president of the working group. "Since we are a global network of varied experiments, we hope the new name will remove negative stereotypes about the type of people who work at the Organization," he adds, pushing up his glasses.

NERDS was chosen by the working group out of a list of names put forward by the CERN community. Among the high contenders were A Large International Experimental Network (ALIEN), the High Energy Laboratory for Physics (HELP), and the initial frontrunner, the Global Organization for Discovery (GOD). However, following a spirited debate in the working group, GOD was discarded on account of unfortunate echoes of the "God particle" – the controversial name accidentally given to the Higgs boson in 1993.

NERDS looks forward to continuing to be at the forefront of scientific research for the next 70 years and beyond.

Naomi Dinmore

Computer Security: Swipes vs PINs vs passwords vs you

What kind of person are you? An artist, like a painter? A credit card fanatic or just "in numbers"? Cerebral, a memoriser or even a genius? An influencer, like a peacock, or just prettily self-confident? A security buff or sufficiently security aware? Or just ignorant about security and your privacy? Let's assume for a moment that the way you unlock your smartphone tells us which.

There are many different ways to unlock your smartphone: swiping patterns, PIN numbers, passwords, biometric fingerprints or face recognition. Some are more secure, some less so. But all are better than nothing. So, let's look at some of them.

Swiping patterns: The obvious choice on Android phones. Your favourite pattern on a 3x3 matrix. But as it should be a continuous swipe, the number of actual possibilities are quite limited, boiling down to about 20 most-used swipes. If yours is listed there, it may be time to move to another, more secure swipe. In any case, your swiping can be spied on and then tried once your smartphone is stolen.
Worse – although it's probably still academic – a small basic sonar system combining a local loudspeaker to emit acoustic signals inaudible to humans and a microphone to record them coming back again allowed researchers to use "the echo signal […] to profile user interaction with the device", i.e. the way your finger swipes over and interacts with the screen. They've shown how this sonar can be employed to help identify the swipe pattern to unlock an Android phone – reducing the number of trials to be performed by an attacker by 70%. And that's only their proof of concept… Maybe PINs and passwords are better?

PINs vs passwords: A common paradigm of computer security is linked to password complexity. Four-digit PIN numbers are no longer state of the art. And even six digits are not necessarily sufficient. While guessing and brute-forcing is difficult, as your smartphone should have a lock-out procedure only allowing a small number of tries before introducing timeouts or even wiping your phone completely(!), PINs can be easily spied on and replayed once your smartphone has been stolen*. Or do you shield your screen as you type your smartphone PIN as you do for your credit card at an ATM? Of course, a better choice is a long and complex password or even passphrase (unless you use one of the top 10 most-used passwords). Admittedly, typing such long and complex passwords can be tedious. Enter: biometrics.

Biometrics: Still our favourite – using your fingerprint sensor or a capture of your face to unlock your phone. Your smartphone (and laptop) manufacturers went to extreme lengths to ensure that your biometric signature cannot be tampered with by your fingerprint on a piece of tape, your face in a photo or your sleeping self. And they also ensured that your biometric information is properly and securely stored using a special-purpose hardware chip (TPM: "trusted platform module"). Still, fingerprint authentication in particular has been broken into in the past for Android and Windows devices, making face recognition our favourite choice to protect access to your smartphone and all the personal (and professional!) data you store and access with it.

*Actually, Apple's latest security patch also fixed some issues with this.

Computer Security Team

Official News

Members of the personnel shall be deemed to have taken note of official news published for the CERN community. Reproduction of all or part of this information by persons or institutions external to the Organization requires the prior approval of the CERN Management. The full list of official news is available here: https://cern.ch/official-news.

Taxation in France

Memorandum concerning the internal tax annual certificate and individual annual statement for 2023 (for the 2024 declaration of 2023 income in France)

The Organization would like to remind members of the personnel that they must comply with the national legislation applicable to them (cf. Article S V 2.02 of the Staff Rules).

I - Internal tax annual certificate and individual annual statement for 2023

The internal tax annual certificate or the individual annual statement for 2023, issued by the Finance and Administrative Processes Department, is available since 12 February 2024 via MyFiles (under "Financial and Social Benefits"). It is intended exclusively for the tax authorities.

1. If you are currently a member of the CERN personnel, you will have received an e-mail containing a link to your certificate or statement, which you can print if necessary.
2. If you are no longer a member of the CERN personnel or are unable to access your certificate or statement as indicated above, you will find information explaining how to obtain one on this page.

II - 2024 tax declaration form of 2023 income in France

The 2024 tax declaration form for 2023 income must be completed following the general indications available on this page.

If you have any specific questions, please contact your local "Service des impôts des particuliers" (SIP, Private Citizens' Tax Office) directly.

HR-Internal-tax@cern.ch

HR department

Exchange rate for the tax declaration form of 2023 income: for the attention of members of the personnel and pensioners living in France

For the tax declaration form of 2023 income, the average annual exchange rate to be used is **EUR 1,05 * for CHF 1**.

**Communicated by the French Tax Authorities*.

HR department

One-to-one meetings with the French tax authorities for CERN members of personnel

In order to help members of the personnel who may need assistance, the Organization has decided to set up individual tax consultation sessions with the French tax authorities (*Service des Impôts des Particuliers (SIP) de Valserhône*) to answer questions on income tax matters in France. These consultations will take place on the Meyrin site, on Friday 26 April 2024 and will be run by six representatives of the SIP. The sessions will require pre-booking an appointment via this portal.

There are a limited number of time slots available, so we recommend you sign up early.

Please note that these meetings will be in French. Should you need any assistance with the language, we advise you to ask a French-speaking colleague for help.

For any questions regarding these meetings, please contact SIPTax.support@cern.ch

HR department

Pension Fund Benefits service: relocation from 22 April to 31 August

Due to the refurbishment of the Pension Fund offices (building 5, 5th floor), please note that, from **22 April to 31 August 2024**, the Benefits service will be relocated to room 510/R-013 (next to the main building).

During this period, meetings will be **by appointment** only.
Thank you for your understanding.

Benefits Service
(pension-benefits@cern.ch or +41 22 767 88 11)

Joint Advisory Appeals Board

The Joint Advisory Appeals Board examined an internal appeal lodged by a staff member against the decision to qualify their performance as “fair” for the reference year 2020. This procedure was suspended pending the decision at the outcome of a harassment investigation procedure. The internal appeal procedure resumed in January 2023.
In application of Article R VI 1.18 of the Staff Regulations, and in agreement with the appellant, the final decision of the Director-General, dated 16 November 2023, and the report of the Joint Advisory Appeals Board are being brought to the attention of the members of personnel.
These documents will be available from 15 April 2024 until 3 May 2024 via this link: https://indico.cern.ch/event/1405017/

HR department

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

16–20 Apr | Knowledge Sharing | Balexert | Geneva International | EN & FR
CERN and International Geneva at Balexert / Le CERN et la Genève internationale à Balexert

18 Apr 19:30 | Knowledge Sharing | CERN Science Gateway | CERN70 | EN
CERN70 public event: The virtuous circle of knowledge and innovation (more CERN70 events)

19 Apr 11:00 | Knowledge Sharing | Council Chamber | Knowledge Transfer & Detector Seminar | EN
Latest development on spectroscopic X-ray imaging in medicine

24 Apr 18:00 | Knowledge Sharing | Le Globe de la science et de l'innovation | Événements publics | FR
Session de présentation : Progrès de l'étude de faisabilité FCC
Cet événement public est destiné aux résidents des communautés locales (l'inscription est nécessaire, le nombre de places étant limité). Nous invitons la communauté du CERN à suivre l'événement en ligne (sans inscription).

25 Apr 18:00 | Knowledge Sharing | Online | CERN Alumni | EN
News from the Lab – CERN Venture Connect

26 Apr 11:30 | At CERN | CERN | EN
Live stream of the funeral of Peter Higgs

26 Apr 14:30 | Knowledge Sharing | Online | CERN Alumni | EN
Moving Out of Academia to Machine Learning

29–30 Apr | Knowledge Sharing | 4/3-006 | Gender in High Energy Theory | EN
GenHET meeting in String Theory

Registration now open

May & Jun | At CERN | Bike to Work | EN & FR
Bike to Work 2024

11 May 19:30 | At CERN | Main Auditorium | CERN MusiClub | EN
Eurovision Song Contest 2024 apèro, karaoke, games and live screening

15 May | Knowledge Sharing | Lausanne | Famelab | EN, FR or DE
Famelab Switzerland local heats

23 May 14:00 | Knowledge Sharing | Online | HR Learning and Development | EN
L&D Micro-talk 10 – Artificial Intelligence (AI) & The Future of Work

2–15 Jun | Accelerators | Sint-Michielsgestel (NL) | CERN Accelerator School | EN
Mechanical & Materials Engineering for Particle Accelerators and Detectors

10–12 Jun | Knowledge Sharing | Online | Sustainable HEP | EN
Sustainable HEP 2024 – 3rd Edition

11–13 Jun | Engineering | CERN | FDF | EN
1st FPGA (Field Programmable Gate Arrays) Developers' Forum meeting

12–25 Jun | Physics | Nakhon Pathom (TH) | Asia-Europe-Pacific School of High-Energy Physics (AEPSHEP) | EN
2024 Asia-Europe-Pacific School of High-Energy Physics

23 Jul–1 Aug | Knowledge Sharing | European Scientific Institute (FR) | I.FAST | EN
Accelerators for healthcare? 10-day innovation challenge

22 Sep–5 Oct | Accelerators | Santa Susanna (ES) | CERN Accelerator School | EN
Introduction to Accelerator Physics

This is a curated list of events relevant to the CERN community. More events are available here: home.cern/events
Indico also shows ALL events happening today, this week and in a calendar view.
If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch

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The March/April issue of the CERN Courier is out

The mid-term report of the Future Circular Collider (FCC) feasibility study demonstrates significant progress across all project deliverables, including physics opportunities, the placement and implementation of the ring, civil engineering, technical infrastructure, accelerators, detectors and cost (pp25–38). While further work is to be done, no technical showstoppers have been found. From the perspective of the technical schedule alone, operations of the Higgs, electroweak and top factory FCC-ee could start in the early 2040s.
In China, a technical design report for the Circular Electron–Positron Collider demonstrates significant progress towards a facility that could operate in the second half of the 2030s, along with excellent prospects for approval and funding (p39). Meanwhile in the US, there is renewed interest in colliding muons, for which major technology challenges remain (p45).
All proposed future colliders have a rich physics case rooted in deeper investigation of the Higgs sector, and Europe needs to move fast to minimise the gap between the LHC and the next collider at CERN (p43). If the feasibility study shows that the FCC can be afforded, then it represents the best available option for the future of CERN and particle physics.
Read the digital edition of this new issue on CDS.

Mentoring@CERN: A new and unique mentoring programme for the CERN community

The LHC Early Career Mentoring Committee and **CERN's** Women in Technology **group** have teamed up to launch a new and unique mentoring programme: **Mentoring@CERN**.

The pilot phase of this initiative, which aims to promote professional development and support within the CERN community, is set to commence on 15 April.

Who can participate?

If you're **associated with CERN through a contract**, whether as staff, fellow, student, user, PART, or external contractor, you're eligible to join. You don't have to work for the LHC experiments or be a woman to take part.

There are **no age or career level restrictions**, ensuring that anyone can apply to be a mentor or mentee. We carefully pair mentors and mentees on a case-by-case basis with personalised matches.

Timeline for 2024

Registration for the pilot round **opens on Monday 15 April 2024 and closes on Wednesday 15 May**. By the end of May, you'll receive news about your match.

A kick-off meeting will take place in June, where we will share more information about mentoring best practices and helpful tools to jumpstart your mentorship experience.

How to participate?

Applying here will take less than 5 minutes. Simply fill out the registration forms for mentors and/or mentees and attach your CV and a short cover letter. Your application will be carefully reviewed to find the perfect match for you.

Behind the scenes

The WIT mentoring programme, established in 2018, focused on pairing mentors and mentees with a special emphasis on female mentees. Meanwhile, the LHC Early Career Mentoring Programme, launched in 2020, connected mentors and mentees working on various LHC experiments.

Inspired by our shared vision, we are combining forces to merge the programmes, uniting our strengths to provide mentorship opportunities to the whole CERN community.

Information session

Whether you want to share your expertise or seek guidance to navigate your career path, we hope Mentoring@CERN will offer a unique experience.

If you are not sure what the programme can provide or if have questions you would like to ask, we invite you to attend our information session **on 25 April at 12:30 CET**: https://indico.cern.ch/event/1403801/

Ready, steady, cycle!

Bike to Work, the Switzerland-wide cycling campaign, is back for its 2024 edition

Join this year's Bike to Work campaign, which encourages workers in companies all over Switzerland to commute by bicycle as often as possible throughout the months of May and June.

Taking part is easy: first form a team of four, give your team a name and register it on the Bike to Work website. No team? No problem: you can also request to join an incomplete team.

There are no registration fees or minimum distance requirements, and part of your journey can be undertaken by public transport. Non-cyclists are not left out: one team member is allowed to commute on foot, by skateboard or by any other means of non-motorised transport.

This year, CERN's participation in the Bike to Work challenge takes on a new dimension. We will be participating alongside the Canton of Geneva and more than 40 other local entities, including municipalities, autonomous public establishments and other international organisations in Geneva.

This means several tens of thousands of us will be coming to work by bike, which will send a strong

and positive message in favour of active mobility, physical activity and good health in Geneva. The "big winners" will be rewarded with a special prize offered by the Canton of Geneva.
You can find more detailed information about the general event and our very own "Bike to CERN" initiative, which runs all year round, on the Bike to Work and Bike to CERN webpages.

Safety first: don't forget to check out the Safety rules for cycling and the Road safety pages and to complete the online course Road Traffic – Bike Riding before getting in the saddle.
May the sun shine and the roads be safe for you all!

Entrance C: amended opening hours

Following damages caused by a car accident on Wednesday 10 April in the morning, Entrance C on the Meyrin site (Route Maxwell) will remain closed for vehicles until Monday 15 April. As of Monday morning, the entrance will be open again for **outgoing vehicles during the usual hours from 7 am to 8 p.m.**, and for **entering vehicles between 7:00 am and 3:00 pm** with a security guard keeping the gate.
The entrance remains accessible to pedestrians and bicycles.
Thank you for your understanding.

SCE department

Obituaries

CERN pays tribute to Peter Higgs

Peter Higgs passed away on 8 April at the age of 94

Peter Higgs, in front of the CMS detector, in 2008. (Image: Maximilien Brice/CERN)

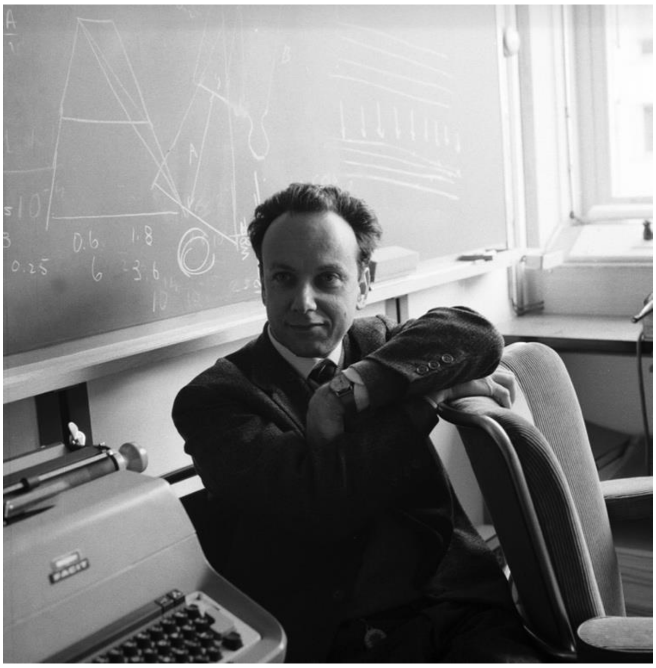
Peter Higgs has passed away at the age of 94. An iconic figure in modern science, Higgs in 1964 postulated the existence of the eponymous Higgs boson. Its discovery at CERN in 2012 was the crowning achievement of the Standard Model (SM) of particle physics – a remarkable theory which explains the visible universe at the most fundamental level.
Alongside Robert Brout and François Englert, and building on the work of a generation of physicists, Higgs postulated the existence of the Brout-Englert-Higgs (BEH) field. Alone among known fundamental fields, the BEH field is "turned on" throughout the universe, rather than flickering in and out of existence and remaining localized. Its existence allowed matter to form in the early universe some 10^{-11} s after the Big Bang, thanks to the interactions between elementary particles (such as electrons and quarks) and the ever-present BEH field. Higgs and Englert were awarded the Nobel prize for physics in 2013 in recognition of these achievements.
"Besides his outstanding contributions to particle physics, Peter was a very special person, an immensely inspiring figure for physicists around the world, a man of rare modesty, a great teacher

and someone who explained physics in a very simple yet profound way," said CERN's Director-General Fabiola Gianotti, expressing the emotion felt by the physics community upon his loss. "An important piece of CERN's history and accomplishments is linked to him. I am very saddened, and I will miss him sorely."

Peter Higgs' scientific legacy will extend far beyond the scope of current discoveries. The Higgs boson – the observable "excitation" of the BEH field which he was the first to identify – is linked to some of most intriguing and crucial outstanding questions in fundamental physics. This still quite mysterious particle therefore represents a uniquely promising portal to physics beyond the SM. Since discovering it in 2012, the ATLAS and CMS collaborations have already made impressive progress in constraining its properties – a painstaking scientific study which will form a central plank of research at the LHC, high-luminosity LHC and future colliders for decades to come, promising insights into the many unanswered questions in fundamental science.

The funeral on 26 April at 11:30 a.m. CET is being live streamed at CERN.

Giuseppe Fidecaro (1926 – 2024)



Giuseppe Fidecaro. (Image: CERN)

We are saddened to learn that experimental physicist Giuseppe Fidecaro, who joined CERN in 1956, passed away on 28 March. He was 97 years old. Giuseppe was a familiar face to the CERN community, often seen arm in arm with his wife Maria as the pair made their way through the CERN corridors. He was also known to CERN visitors, featuring prominently in the film shown in the Synchrocyclotron exhibition. Maria Fidecaro passed away in September 2023.

Born in Messina, Italy, in 1926, Giuseppe studied physics at the University of Rome under the supervision of Edoardo Amaldi , graduating in 1947. He came to CERN with Maria in the summer of 1956 and was assigned to the Synchrocyclotron. There, he established a group and prepared equipment for experiments which, in 1958, enabled the successful search for pions decaying into an electron and a neutrino – a process that helped to support the existence of the universal Fermi interaction.

In 1975, Giuseppe was appointed as co-chair of a joint scientific committee set up under a collaboration agreement between CERN and the former USSR concerning the use of atomic energy, a responsibility he held until 1986. He was also tasked with coordinating cooperation with the Joint Institute for Nuclear Research in Dubna. Giuseppe officially retired in 1991 but, together with Maria, continued his work at CERN as an honorary member of the personnel until as recently as 2020.

A funeral service was held at the Italian Catholic Mission of Geneva at 10.00 a.m. on Friday, 5 April.

A full obituary will appear in the CERN Courier.

Ombud's corner

Difficult conversations (part II): "The feelings"

In the first part of this three-part article , I explored with you how to tackle the most obvious aspect of any "difficult conversation": the tip of the iceberg, the question of what happened?

In this second part, we come back to Peter*, Jenny* and Andrzej*, who are still preparing for the conversations they dread.

In their cases, as in all difficult conversations, **feelings lie at the heart of what is wrong** and are the primary issues at stake:

> Peter feels furious that Michael has started to criticise him openly.
>
> Jenny feels irritated having to cope with Mona's constant unacceptable behaviour.
>
> Andrzej feels worried when his thesis supervisor, Tom, fails to respond to his request for more involvement and feedback.

Most of us find it quite difficult to express our feelings but, ignoring what is behind the issues we're facing will not help us to resolve them, as these feelings will re-surface sooner or later and create other issues. If feelings are the real issue, then feelings should be addressed.

The first difficulty we meet when expressing our feelings **is avoiding making a judgement and/or an assumption.** It's easy to spot when someone falls into this pitfall: if someone says "**I feel like** … "then what follows is most probably not a feeling but an accusation that will not help the conversation.

> If Peter says: "I feel like you are trying to isolate me from the rest of the team", Peter is not expressing his feelings but merely making an assumption of why Michael is behaving this way.
>
> If Jenny says "I feel like you are judging the competence of your colleagues ", Jenny is assuming that this is Mona's intention.
>
> Similarly, if Andrzej says "I feel like my work is of no importance to you ", he is assuming that Tom has no interest in his thesis, and he is also making a judgement that Tom is not an effective supervisor.

If you can avoid this trap of confusing feelings with judgements and assumptions, and if you express genuine feelings, no one can deny them or object to them because **your feelings belong to you**.

Bringing feelings into the conversation is an essential step in a difficult conversation, but it is far from easy, and caution and honesty must be exercised to get it right.

Very often, we may focus on a single feeling such as anger, disappointment or disgust, but it is very important to **see the whole range of our feelings before we talk about them**. Are we just angry or are we also overwhelmed? Are we simply disappointed or also very worried? Are we just disgusted or are we also wholly disheartened about a conflictual situation in the workplace?

> Peter finds himself absolutely furious with Michael. However, when Peter takes a minute to think beyond his anger, he realises that he also feels resentful, disheartened and humiliated being isolated from the rest of the team.
>
> Jenny's main feeling may seem to be sheer irritation. But, thinking twice about it, she may discover that she is also exasperated and outraged by Mona's disparaging comments about other team members.
>
> Andrzej is extremely worried by his supervisor's lack of feedback, but he realises that he is also highly discouraged and feeling very insecure.

Once we have identified what our feelings are, in all their complexity, we also need to be honest about them and "negotiate" with them. Where do these feelings come from? **What is the story we are telling ourselves** that leads to these feelings? What could be the feelings of the other party? Very often, an **increased awareness of the other person's story changes how we feel.**

> Reflecting on why he is feeling so furious with Michael, Peter may realise that this is not the first time he's faced open opposition from a colleague and that this has already caused damage to his career before. If, instead of dwelling on the past,

Peter sticks to what the issue with Michael is today, his feelings of resentment and disheartenment might lessen and take on more reasonable proportions.

Finally, a key part of the “feelings “conversation is that **feelings should be acknowledged.** In a conflictual situation, you cannot ensure that the other party will acknowledge your feelings, but you can show the way by acknowledging their feelings. This is particularly important if you are higher in the hierarchy or have more power than the other party.

If Mona, in response to Jenny’s reproaches, shares her feelings of insecurity and disconnection from the other team members, Jenny should acknowledge these feelings:” It sounds like you are really upset about this.”

If Andrzej rakes up the courage to tell his thesis supervisor, Tom, how he is genuinely suffering from the lack of proper supervision, Tom should acknowledge these feelings: “This seems really important to you. If I were in your shoes, I’d probably feel discouraged too.” This will place the conversation on a much more constructive footing.

When considering whether or not to have a difficult conversation and, later on, when preparing for it, keep in mind that difficult conversations do not just involve feelings but are about feelings. You may want to think about your feelings, all of them, where they come from and how you can influence them. Prepare to acknowledge the feelings of the other party.

This “feelings“ conversation will help you turn a difficult conversation into a learning conversation with the potential to preserve and even improve the working relationship. Remember, the CERN Ombud can help you to prepare for a difficult conversation.

In the next and final article on the topic of “difficult conversations”, I propose to explore with you the third entangled part of such conversations, the “identity” conversation, in other words, how a conversation may pose a threat to the story we are telling ourselves about ourselves.

Laure Esteveny

** Names and situations are fictitious.*

This article is inspired by the book *Difficult Conversations* by Douglas Stone, Bruce Patton and Sheila Heen, which I strongly recommend. It is available to loan from the CERN Library.

I would like to hear your reactions and suggestions – join the CERN Ombud Mattermost team at https://mattermost.web.cern.ch/cern-ombud/